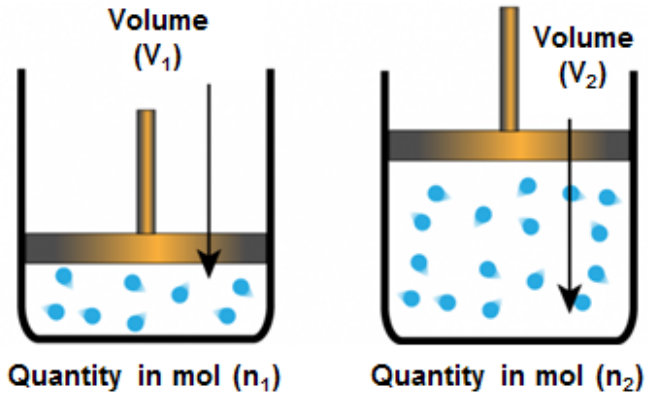
The 4 Laws of Ideal Gas

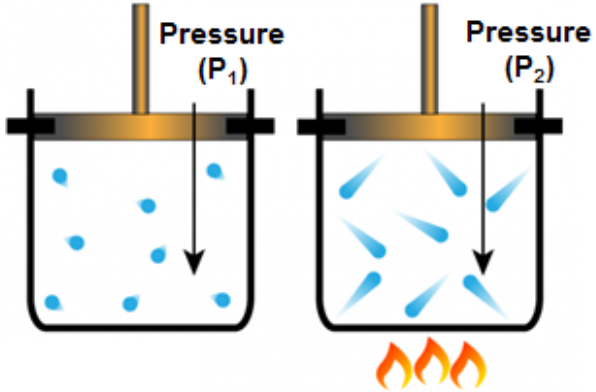
THE AVOGADRO'S LAW
(of variable volume-V & amount of gas-n)

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$



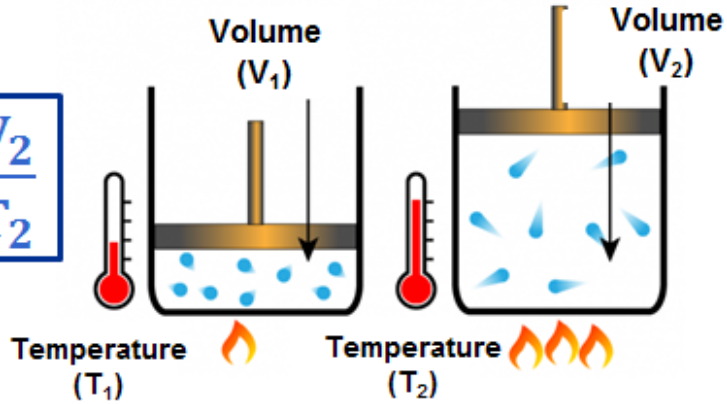
THE GAY-LUSSAC'S LAW
(of variable pressure-P & temperature-T)

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$



THE CHARLES'S LAW
(of variable volume-V & temperature-T)

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$



THE BOYLE'S LAW
(of variable pressure-P & volume-V)

$$P_1 V_1 = P_2 V_2$$

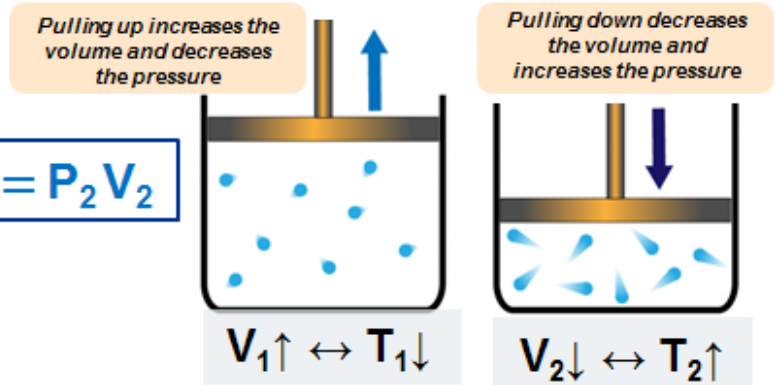
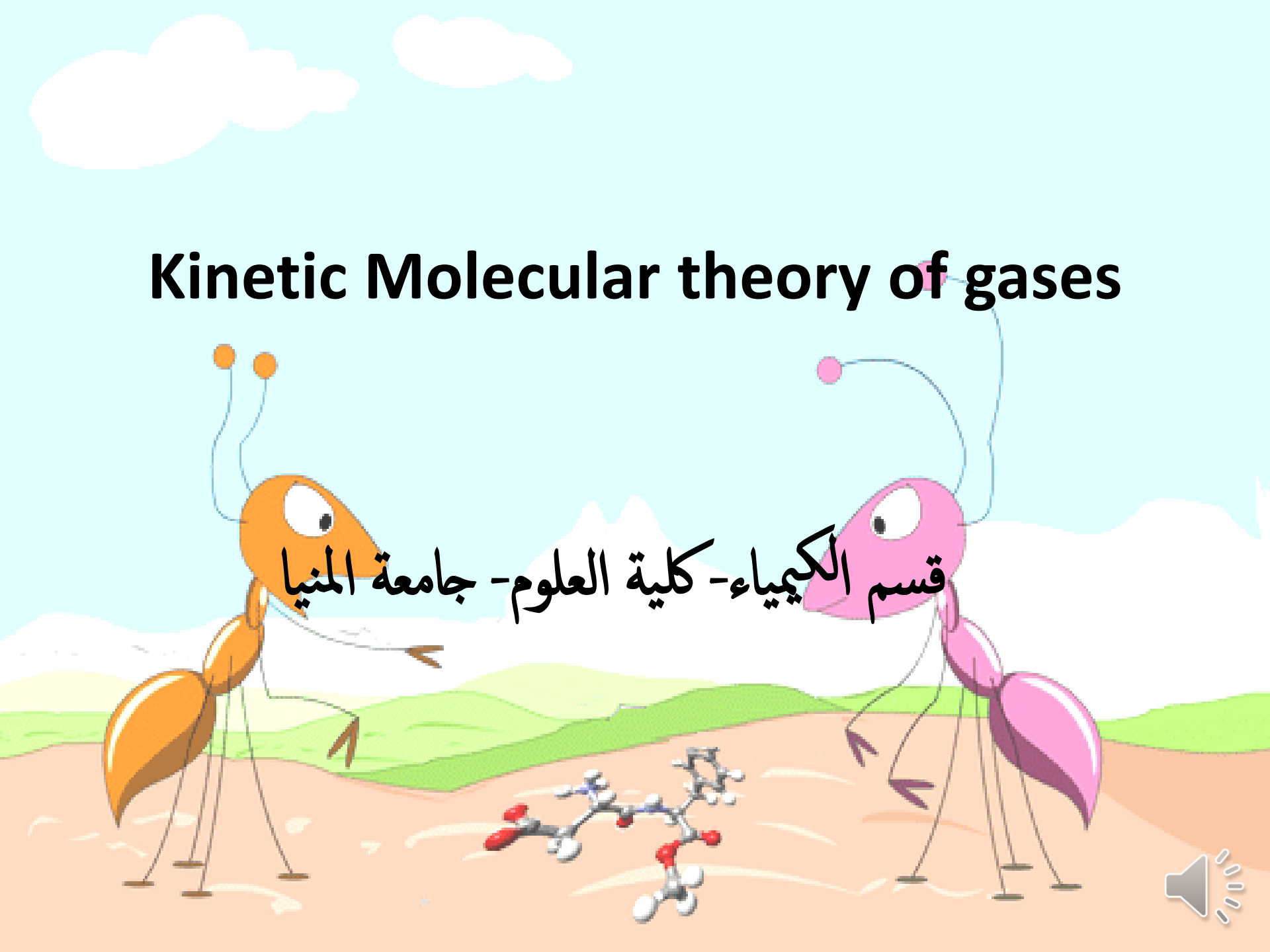


FIGURE 1

Kinetic Molecular theory of gases



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Postulates

The basic assumptions of kinetic molecular theory are:

❑ Postulate 1

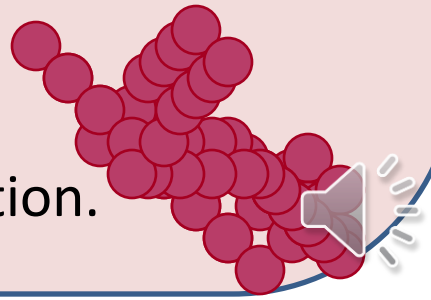
- Gases consist of discrete molecules that are relatively far apart.
- Gases have few intermolecular attractions.
- The volume of individual molecules is very small compared to the gas's volume

❖ **Evidence :** Gases are easily compressible.

❑ Postulate 2

- Gas molecules are in constant, random, straight-line motion with varying velocities.

❖ **Evidence** : Brownian motion displays molecular motion.



Postulates

❑ **Postulate 3**

- Gas molecules have elastic collisions with themselves and the container.
- Total energy is conserved during a collision.

❖ **Evidence**: A sealed, confined gas exhibits no pressure drop over time.

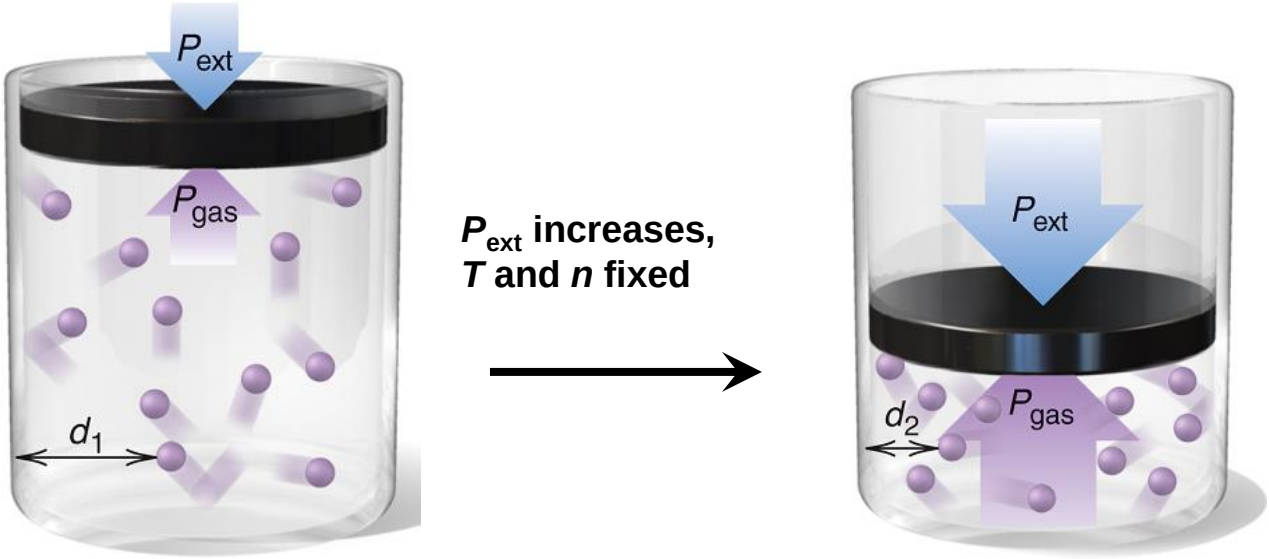
❑ **Postulate 4**

- The kinetic energy of the molecules is proportional to the absolute temperature.
- The average kinetic energies of molecules of different gases are equal at a given temperature.

❖ **Evidence:** Brownian motion increases as temperature increases.

Gas Laws Explained – Boyle's Law

- Boyle's Law says that the volume of a gas is inversely proportional to the pressure
  - ✓ Decreasing the volume forces the molecules into a smaller space.
- More molecules will collide with the container at any one instant, increasing the pressure.

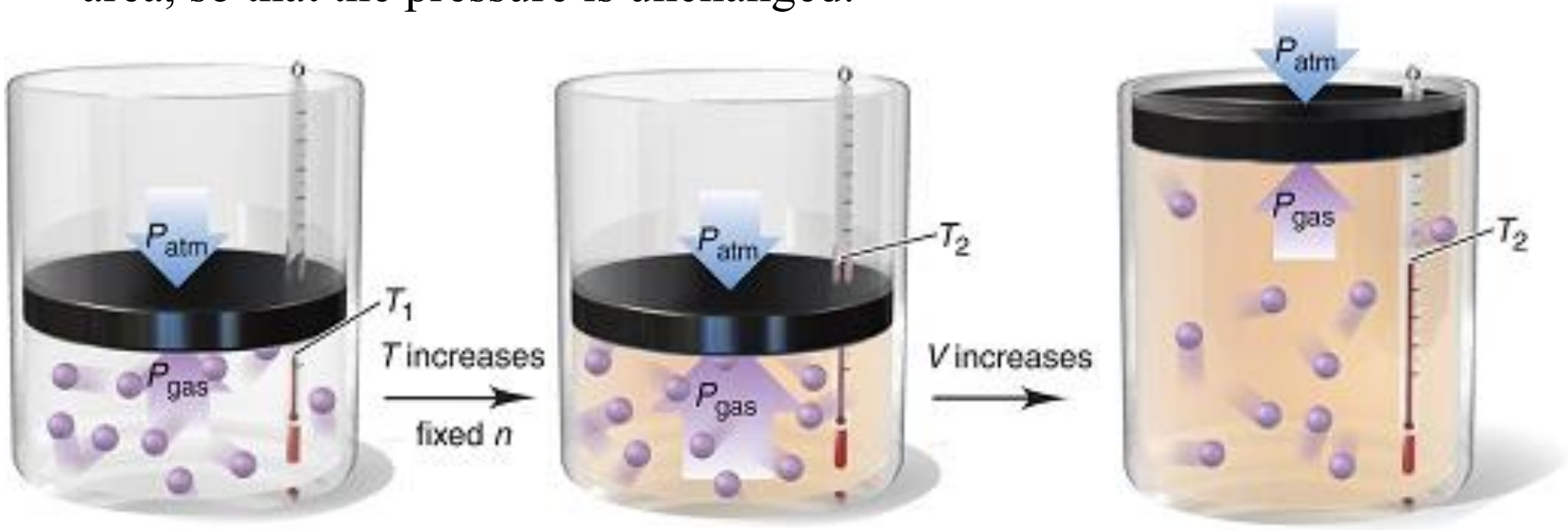


At any T, $P_{gas} = P_{ext}$ as particles hit the walls from an average distance, d_1.

Higher P_{ext} causes lower V, which results in more collisions, because particles hit the walls from a shorter average distance ($d_2 < d_1$). As a result, $P_{gas} = P_{ext}$ again.

Gas Laws Explained – Charles's Law

- Charles's Law says that the volume of a gas is directly proportional to the absolute temperature.
  - ✓ According to kinetic molecular theory, when we increase the temperature of a gas, the average speed, and thus the average kinetic energy, of the particles increases.
- The greater volume spreads the collisions out over a greater surface area, so that the pressure is unchanged.



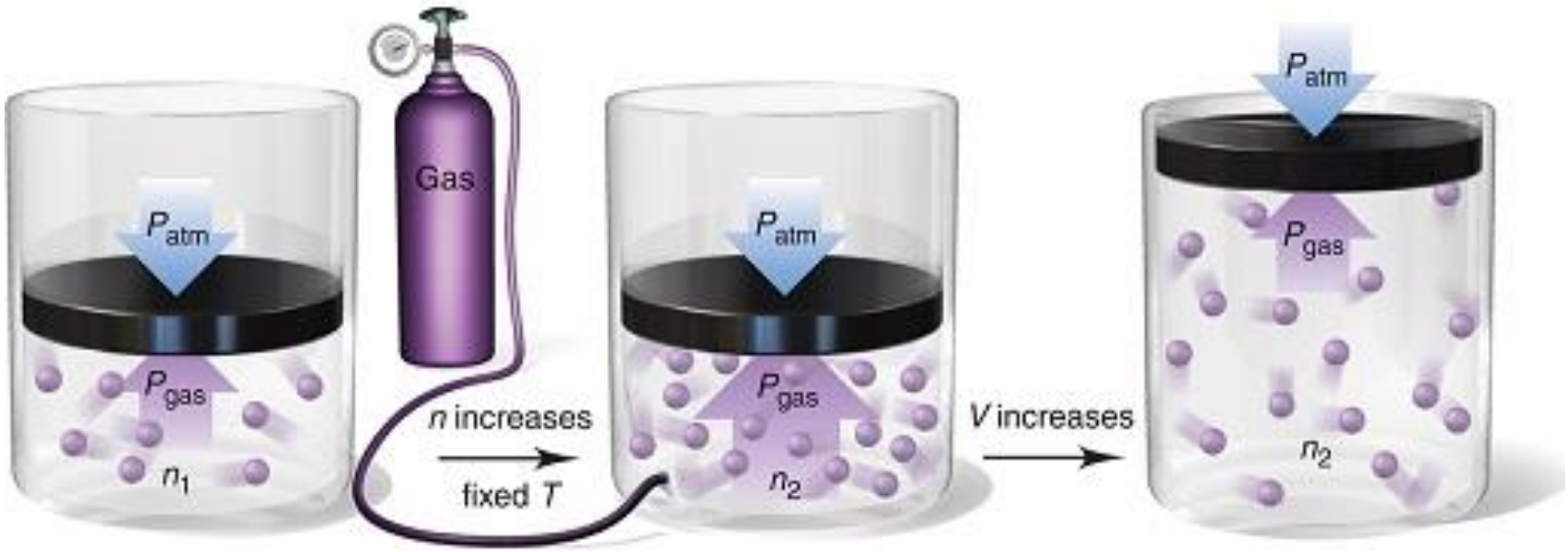
At T_1, $P_{gas} = P_{atm}$.

Higher T increases collision frequency, so $P_{gas} > P_{atm}$.

Thus, V increases until $P_{gas} = P_{atm}$ at T_2.

Gas Laws Explained – Avogadro's Law

- Avogadro's Law says that the volume of a gas is directly proportional to the number of gas molecules.
- Increasing the number of gas molecules causes more of them to hit the wall at the same time.
- To keep the pressure constant, the volume must then increase.



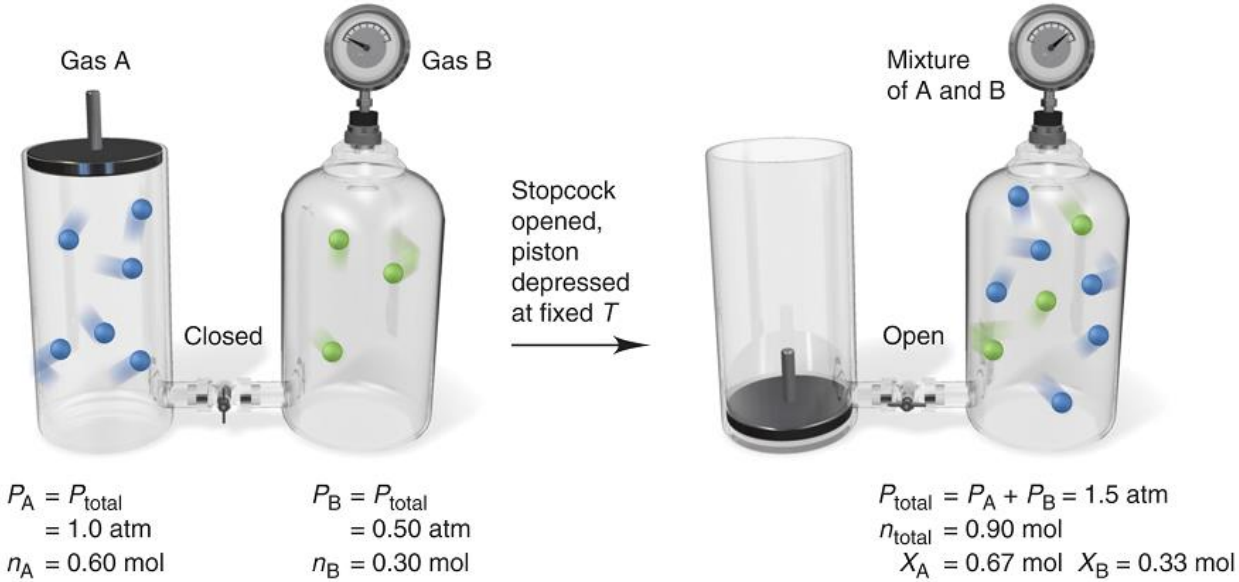
For a given amount, n_1, of gas, $P_{gas} = P_{atm}$.

When gas is added to reach n_2 the collision frequency of the particles increases, so $P_{gas} > P_{atm}$.

As a result, V increases until $P_{gas} = P_{atm}$ again.

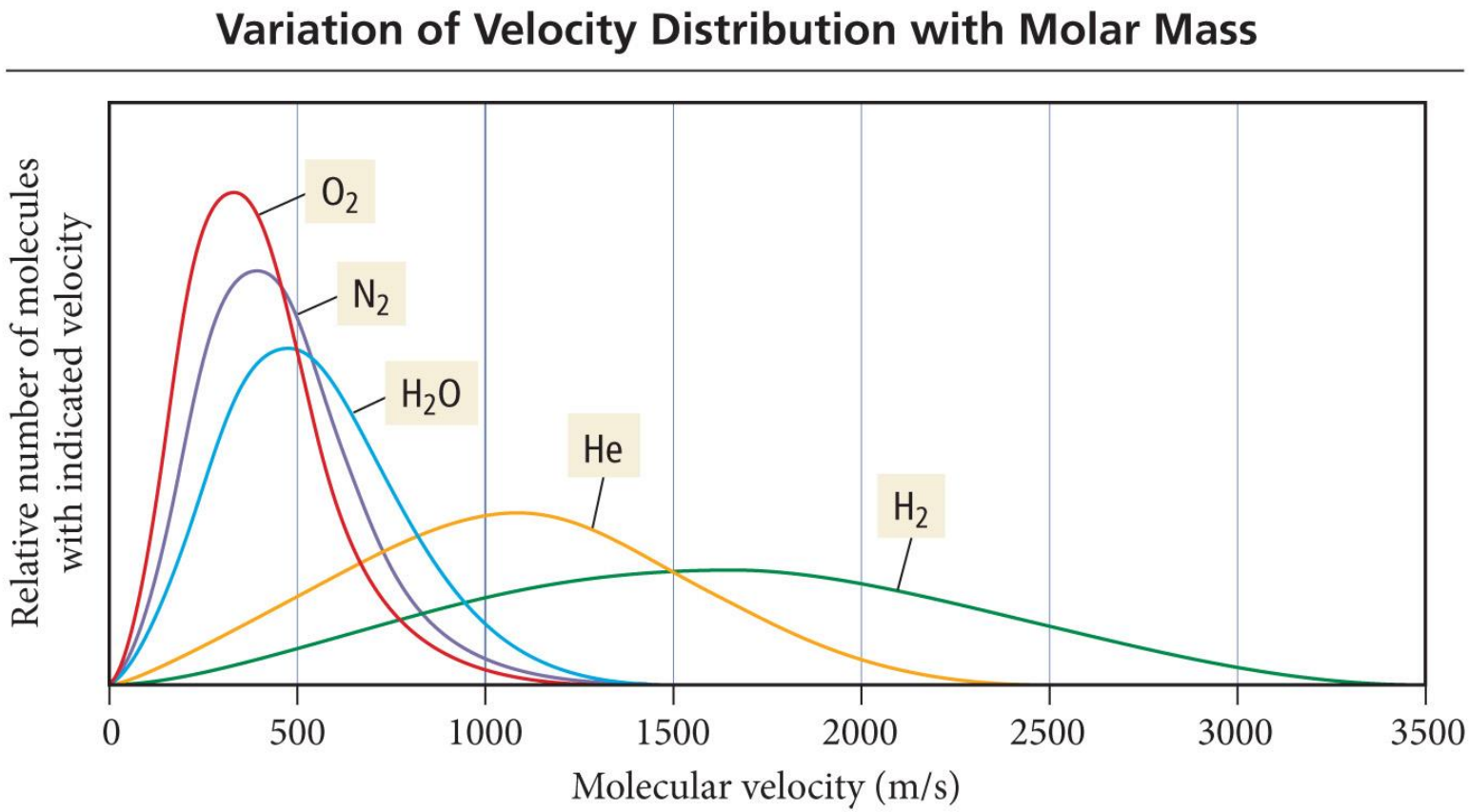
Gas Laws Explained – Dalton's Law

- Dalton's law: the total pressure of a gas mixture is the sum of the partial pressures.
- According to kinetic molecular theory, the particles have negligible size and they do not interact.
  - ✓ Particles of different masses have the same average kinetic energy at a given temperature.
- Because the average kinetic energy is the same, the total pressure of the collisions is the same.



Molecular Speed versus Molar Mass

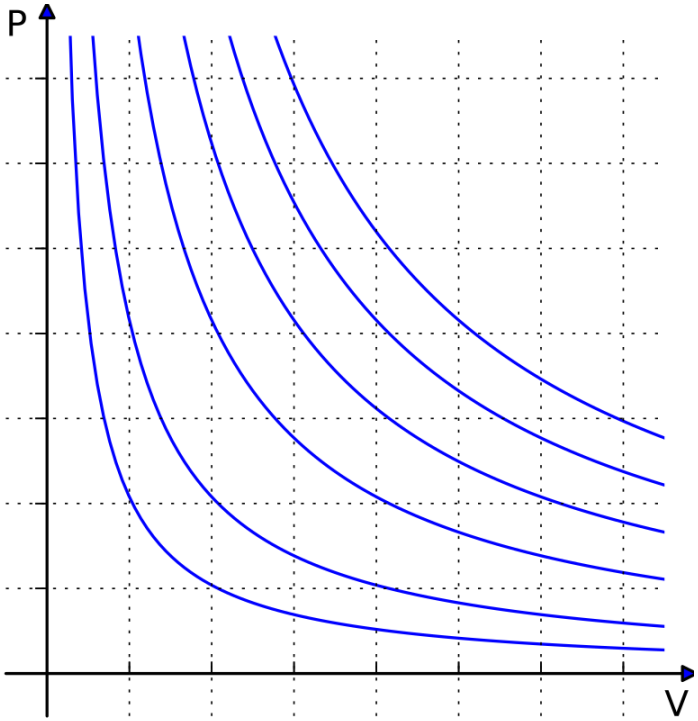
- To have the same average kinetic energy, heavier molecules must have a slower average speed.



Ideal Gas Law

- The ideal gas law, also called the general gas equation, is the equation of state of a hypothetical ideal gas.
- It was first stated by ***Benoît Paul Émile Clapeyron*** in 1834 as a *combination of the empirical Boyle's law, Charles's law, Avogadro's law, and Gay-Lussac's law.*

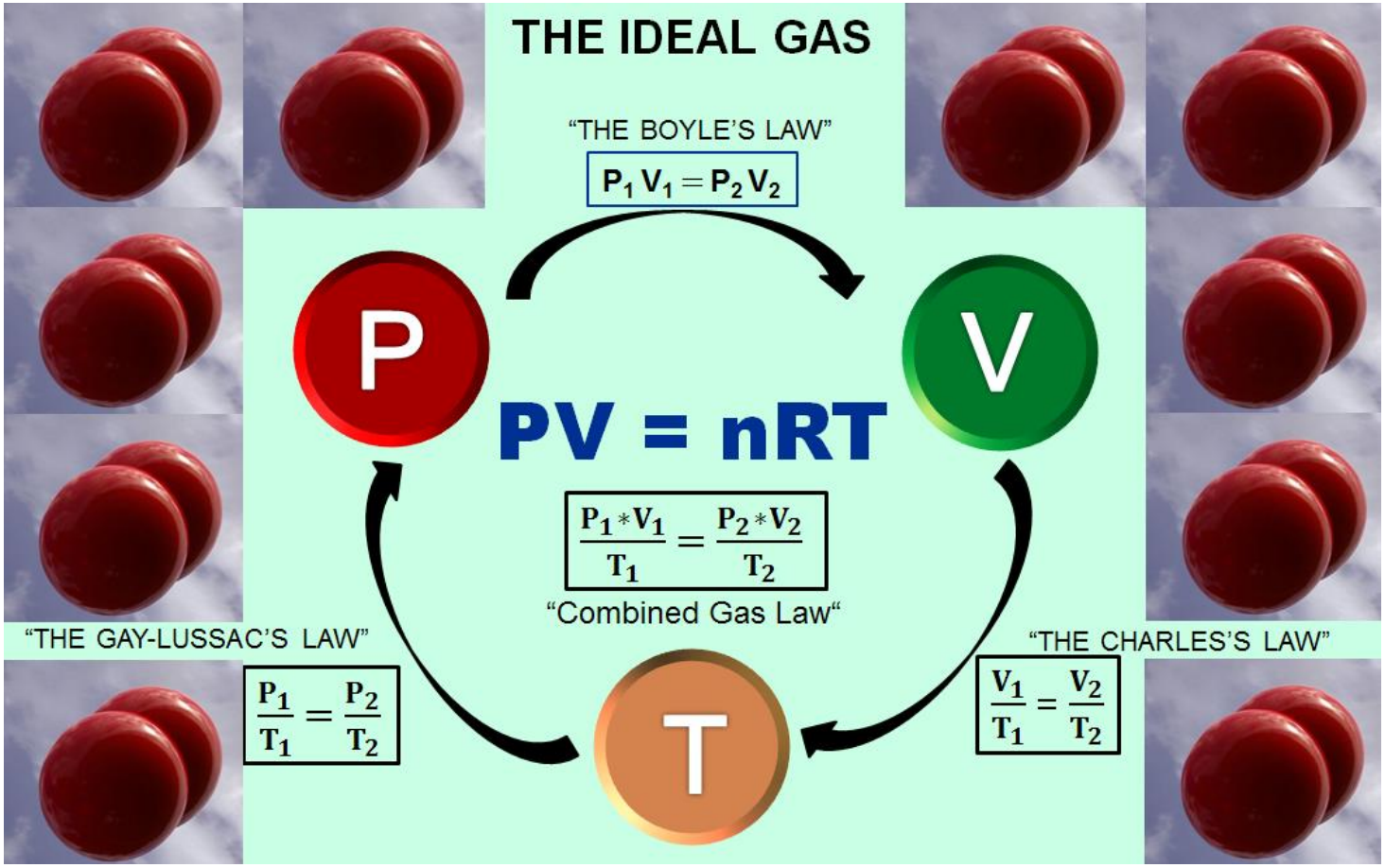
- Isotherms of an ideal gas for different temperatures.
- The curved lines are rectangular hyperbolae



Ideal Gas Law

$$PV = nRT$$

1. P is pressure measured in atmosphere
2. V is volume measured in Liters
3. n is moles of gas present
4. R is a constant that converts the units. Its value is 0.0821 atm.L/mol.K
5. T is temperature measured in Kelvin
6. Simple algebra can be used to solve for any of these values.

$$p = \frac{nRT}{V} \quad V = \frac{nRT}{P} \quad n = \frac{PV}{RT} \quad T = \frac{PV}{nR} \quad R = \frac{PV}{nT}$$

Ideal Gas Law